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AgriVision: A Data Science–Driven Sustainable Agricultural Analytics and Visualization Platform

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0405- FUNDAMENTALS OF DATA SCIENCE

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING (AI)**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “AgriVision: A Data Science–Driven Sustainable Agricultural Analytics and Visualization Platform” has been carried out by S.Sirivally (192472371) under the supervision of Dr. KrishnaRaj Mony. and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech. Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I , S.Sirivally of the CSE-AI , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “AgriVision: A Data Science–Driven Sustainable Agricultural Analytics and Visualization Platform” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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(Mandatory for all team projects)

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ABSTRACT

Agriculture plays a vital role in food security, economic development, and sustainable resource management. However, agricultural datasets often contain large volumes of heterogeneous information related to crops, soil, rainfall, temperature, irrigation, fertilizer usage, weather conditions, and crop yield. Analysing such datasets manually can be time-consuming and may make it difficult to identify important patterns and relationships. AgriVision is a web-based agricultural data analysis and visualization system developed to simplify the process of uploading, preprocessing, analysing, visualizing, and interpreting agricultural datasets.

The proposed system provides an integrated platform where users can upload agricultural datasets in a structured format and perform essential preprocessing operations such as data validation, missing-value handling, duplicate detection, and preparation of data for analysis. After preprocessing, the system performs statistical and exploratory analysis to identify important characteristics and trends within the dataset. Various visualizations, including histograms, box plots, scatter plots, heatmaps, and bar charts, are generated to provide a clear graphical representation of agricultural data. These visualizations help users understand data distributions, detect outliers, compare agricultural parameters, and identify relationships between different variables.

AgriVision also provides a centralized dashboard for presenting analysis results in an understandable manner. The system supports report generation, allowing users to view and download summarized analysis results for further reference and documentation. In addition, the system is designed to support agricultural prediction and data-driven decision-making by utilizing relevant agricultural parameters. By integrating data preprocessing, statistical analysis, visualization, dashboard presentation, reporting, and prediction within a single web application, AgriVision reduces the need for users to rely on multiple separate tools.

The system is intended to assist students, researchers, agricultural analysts, and other users in exploring agricultural datasets efficiently. Overall, AgriVision demonstrates how web technologies, data analytics, visualization techniques, and machine learning can be combined to transform raw agricultural data into meaningful and accessible information for better agricultural analysis and decision-making.

Keywords: Agricultural Data Analysis, Crop Yield Prediction, Data Preprocessing, Data Visualization, Exploratory Data Analysis, Machine Learning, Agricultural Datasets, Dashboard, Statistical Analysis, Smart Agriculture.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
CSV	Comma-Separated Values	—
API	Application Programming Interface	—
KPI	Key Performance Indicator	—
DB	Database	—
PDF	Portable Document Format	—
UI	User Interface	—
DPDP Act	Digital Personal Data Protection Act, 2023 (India)	—
FastAPI	Python web-application backend framework used in this project	—
t/ha	Tonnes per hectare (yield unit)	t/ha
mm	Millimetres (rainfall unit)	mm

CHAPTER-1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The rapid growth of agricultural data collection, data analytics, and web-based technologies has created new opportunities for improving the way agricultural information is analysed and presented. Agricultural datasets can contain large amounts of information related to crops, soil conditions, rainfall, temperature, irrigation, fertilizer usage, weather conditions, production, and yield. Presenting the results of such analysis in a clear and understandable manner is important for researchers, students, agricultural analysts, and decision makers. AgriVision – Agricultural Data Analysis and Visualization System is developed as a web-based platform that integrates agricultural data processing, analysis, visualization, dashboard presentation, and report management within a single system.

The system is designed to convert analysed agricultural data into meaningful results through interactive dashboards, statistical summaries, visualizations, and downloadable reports. By providing a centralized interface for managing analysis results, AgriVision reduces the need for users to depend on separate tools for viewing charts, interpreting statistical results, and preparing reports. The system therefore provides a structured approach for presenting agricultural information and supporting data-driven understanding of agricultural datasets.

1.2 Need for the Project

Agricultural datasets can be large and contain multiple variables, making it difficult for users to manually understand and organize the results of data analysis. Even when statistical analysis and visualization are performed successfully, users may face difficulties in accessing the results in a structured form. Preparing reports manually from different charts and statistical outputs can also require additional time and effort. Therefore, there is a need for a centralized system that can present agricultural analysis results through an interactive dashboard and automatically organize important results into reports.

AgriVision addresses this need by providing a dedicated dashboard for presenting important agricultural analysis results and a report management facility for viewing and downloading generated reports. The system enables users to access statistical summaries and visualizations in an organized manner, making the analysis easier to understand and use for further study or decision-making.

1.3 Problem Statement

The project aims to develop an agricultural data analysis and visualization system that can present processed agricultural data and analysis results through an interactive dashboard and structured reports. The system must provide users with clear access to statistical summaries, generated visualizations, analysis results, and reports while maintaining consistency and usability. The system should also support efficient result management so that users can view and download reports generated from the analysed datasets without manually compiling the results.

1.4 Project Aim

The aim of the Dashboard, Reports and Result Management module of AgriVision – Agricultural Data Analysis and Visualization System is to develop a user-friendly interface for presenting, organizing, and

managing the results obtained from agricultural data analysis. The module is designed to display important statistical information and visualizations through an interactive dashboard, provide access to generated reports, and allow users to view and download analysis results. It aims to make complex agricultural analysis outputs easier to understand, access, and maintain within a single web-based platform.



Fig 1: Agricultural Data Analytics Dashboard

1.5 Project Objectives

The primary objective of this module is to develop an interactive dashboard that presents important agricultural analysis results in a clear and organized manner. The module aims to integrate statistical summaries and visualization outputs generated from the analysis process and display them in a user-friendly format. It also aims to provide automated report generation and report management facilities so that users can view and download the results for documentation and future reference. Another objective is to ensure that analysis results are presented consistently across different agricultural datasets while providing simple navigation and easy access to the required information.

1.6 Scope

Category	Description
What is Included	Interactive dashboard, statistical result presentation, visualization display, analysis result management, report generation, report viewing, and report downloading.
What is Excluded	Manual modification of analytical results, independent agricultural decision-making, replacement of professional agricultural expertise, and collection of agricultural data from physical fields.
Assumptions	The uploaded dataset has been successfully processed and analysed by the preceding modules, and the required analysis results and visualizations are available for presentation.

Boundary Conditions	Dashboard and report outputs depend on the quality and availability of processed data, analysis results, generated visualizations, system performance, and supported report formats.
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1.7 Expected Engineering Outcome

The expected engineering outcome is a functional Dashboard, Reports and Result Management module integrated into the AgriVision web application. The module provides an interactive dashboard that presents agricultural analysis results through statistical summaries and graphical visualizations. It provides a structured result management mechanism through which users can access the outputs generated from the analysis process. The module also generates reports containing relevant agricultural analysis information and provides facilities for viewing and downloading the generated reports. The overall process can be represented as Analysis Results → Dashboard Presentation → Result Management → Report Generation → Report Viewing/Downloading. The final outcome is a centralized and user-friendly result management interface that improves the accessibility, organization, and presentation of agricultural analysis results.

CHAPTER-2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research and existing technologies related to agricultural data visualization, dashboard systems, data analytics, decision support systems, and automated reporting were reviewed to understand how agricultural information can be presented effectively. Research literature was considered from academic sources and journals using keywords such as Agricultural Data Visualization, Agricultural Dashboards, Agricultural Decision Support Systems, Crop Data Analysis, Data Analytics in Agriculture, and Agricultural Reporting Systems. The review focused on approaches that transform processed agricultural data into meaningful visual and summarized information that can be accessed by different types of users. The review examined the use of statistical summaries, graphical visualization, interactive dashboards, data management techniques, and automated report generation in agricultural applications. Existing solutions were analysed based on their ability to organize large datasets, communicate analytical results, support user interaction, and provide useful information for decision-making. The findings helped identify the requirements and design considerations for the Dashboard, Reports and Result Management module of AgriVision.

2.2 Review of Existing Work

Existing agricultural information systems commonly use data analytics and visualization techniques to help users understand patterns in crop production, yield, environmental conditions, and other agricultural parameters. Dashboards are increasingly used to provide a centralized view of important indicators and analytical results. Graphical representations such as bar charts, line charts, scatter plots, heatmaps, histograms, and box plots can make large and complex datasets easier to interpret compared with viewing raw tabular data.

Agricultural decision support systems generally combine processed data with analytical outputs to assist users in understanding agricultural conditions and trends. However, many systems are developed for specific agricultural applications, datasets, or user groups. Their dashboards may focus on a limited number of indicators, while report generation and result management may not be integrated into the same platform. This creates a requirement for a more general-purpose system capable of presenting different agricultural datasets and their corresponding analysis results through a common interface.

Automated reporting is another important aspect of agricultural data management. Instead of manually collecting charts, statistics, and observations from different parts of an application, an integrated reporting facility can organize the results into a structured document. Such an approach reduces manual effort and provides a permanent record of the analysis. AgriVision incorporates these concepts by connecting the analysis outputs with an interactive dashboard and report management system.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
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Manual Analysis and Result Preparation	Provides direct control over analysis and presentation	Time-consuming and difficult when datasets are large	Motivates automated result presentation and management
Spreadsheet-Based Agricultural Analysis	Easy to use and supports tables and basic charts	Requires manual creation and organization of charts and reports	Provides the basis for automated visualization
Agricultural Visualization Systems	Makes agricultural patterns easier to understand through graphical representations	Often designed for specific datasets or applications	Supports integration of multiple visualization types in AgriVision
Agricultural Dashboard Systems	Provides centralized access to important indicators and analytical results	Some systems provide limited customization and may focus on specific agricultural parameters	Supports the interactive dashboard developed in AgriVision
Agricultural Decision Support Systems	Helps users interpret data and support data-driven decisions	May require specialized data sources and domain-specific configurations	Provides a foundation for presenting meaningful agricultural insights
Automated Reporting Systems	Reduces manual effort in preparing analysis reports	Reporting may be separated from the analysis and visualization process	Supports integrated report generation and downloading
Proposed AgriVision Module	Combines dashboard, visualization, result management, and reporting	Depends on the availability and quality of processed analysis results	Provides a centralized interface for accessing agricultural analysis outputs

2.4 Research / Engineering Gap

Existing agricultural analysis and visualization solutions demonstrate the usefulness of dashboards and graphical representations for understanding agricultural information. However, there is still a need for an integrated platform where statistical results, visualizations, dashboards, and reports can be accessed through a common interface. In many conventional approaches, users may need to manually organize analysis outputs or use separate applications to prepare reports after completing the analysis.

Another limitation is the lack of a unified result management mechanism for agricultural datasets. When multiple datasets are analysed, users need an organized way to identify the dataset, access its corresponding charts and statistical results, and retrieve the generated report. Without centralized result management, analysis outputs can become difficult to locate and compare. AgriVision addresses this engineering gap by integrating dashboard presentation, visualization access, result management, and report generation into one web-based system.

The proposed module also focuses on usability. Rather than presenting raw analytical outputs directly to users, the system organizes them into a structured dashboard and report format. This allows users to

understand the results more efficiently and provides a convenient mechanism for storing, viewing, and downloading the outputs of agricultural analysis.



Fig 2: Challenges in Traditional Agricultural Data Management

2.5 Proposed Contribution

The Dashboard, Reports and Result Management module contributes an integrated result presentation and management layer to the AgriVision system. The module receives the statistical analysis and visualization outputs generated by the agricultural data analysis process and presents them through an interactive dashboard. Important results are organized so that users can easily navigate between datasets, statistical information, and graphical visualizations.

The module also provides automated report generation and management facilities. Analysis results can be compiled into structured reports that users can view and download for documentation, academic work, further analysis, and future reference. By combining dashboard presentation, visualization access, result management, and reporting within a single application, the proposed module reduces manual effort and improves the accessibility of agricultural analysis results.

The overall contribution can be represented as Processed Agricultural Data → Analysis Results → Dashboard → Visualization and Statistical Results → Report Generation → View/Download → Data-Driven Understanding. This integration provides a practical solution for managing the final outputs of agricultural data analysis in a centralized web-based environment.

CHAPTER-3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The Dashboard, Reports and Result Management module is designed to provide a centralized interface for displaying and managing the outputs generated from agricultural data analysis. The module receives processed analysis results, statistical summaries, and visualization outputs from the analysis module and presents them through an interactive dashboard. It also provides report generation, report viewing, and report downloading facilities. The design focuses on usability, organized presentation, reliable result access, and efficient management of outputs generated for different agricultural datasets.

Stakeholder Requirements

Stakeholder	Requirement
Students	View agricultural analysis results through a simple and understandable dashboard.
Researchers	Access statistical results and visualizations for further study and interpretation.
Agricultural Analysts	View graphical representations and summarized agricultural information in an organized manner.
Users	Select and access results corresponding to the required dataset.
Users	View generated reports containing important analysis results.
Users	Download reports for documentation and future reference.
Project Administrators	Maintain organized access to datasets, analysis outputs, charts, and reports.

Functional and Non-Functional Requirements

Requirement	Type	Measurable Target
Display analysed dataset information	Functional	Correct dataset name and identification displayed
Display statistical summaries	Functional	Relevant statistical results displayed correctly
Display generated visualizations	Functional	Available charts loaded successfully
Dashboard navigation	Functional	Users can access required results with minimal steps
Generate analysis report	Functional	Report generated successfully for processed datasets
View generated report	Functional	Report opens correctly through the application
Download generated report	Functional	Report can be downloaded successfully
Result management	Functional	Results are organized according to the corresponding dataset
Dashboard response time	Non-Functional	Normal dashboard requests completed within a reasonable response time
Usability	Non-Functional	Users can access major functions without complex navigation

Reliability	Non-Functional	Correct results displayed consistently
Data security	Non-Functional	Uploaded data and generated results handled securely
Compatibility	Non-Functional	Application accessible through commonly used web browsers

3.2 Constraints & Applicable Standards

The major constraints of the module include dependency on the results generated by the preceding data processing and analysis modules, availability of visualization files, report generation performance, supported file formats, storage capacity, network conditions, and browser compatibility. If the dataset has not been successfully processed or the required analysis output is unavailable, the dashboard cannot provide the corresponding visualization or report.

The module also needs to ensure that results are displayed accurately and that generated reports correspond to the correct dataset. Since agricultural datasets may contain large numbers of records and multiple variables, efficient storage and retrieval of generated results are important engineering considerations.

Standard / Code	Requirement	How Applied
ISO/IEC 25010	Software quality and usability	Applied through usability, reliability, performance, and maintainability considerations
ISO/IEC 27001	Information security	Secure management of datasets, reports, and system information
OWASP Top 10	Secure web application development	Protection against common web application security risks
WCAG Principles	Accessible web content	Clear presentation, readable text, navigation, and understandable interface
Data Privacy Principles	Protection of user and dataset information	Appropriate handling and restricted access to stored information

3.3 Design Specifications

The design specifications define the expected performance and functional characteristics of the Dashboard, Reports and Result Management module.

Parameter	Target / Requirement	Unit	Priority
Dashboard Loading Time	≤10	seconds	High
Visualization Loading	Successful display of available charts	%	High
Report Generation	Successful generation for processed datasets	%	High
Report Download	Successful download of generated reports	%	High
Result Accuracy	≥95	%	High
System Availability	≥95	%	Medium
User Navigation	Simple and clear navigation	—	High
Data Security	Secure storage and controlled access	—	High

Browser Compatibility	Support for commonly used browsers	—	Medium
Report Accessibility	View and download generated reports	—	High

3.4 Alternative Solutions & Evaluation

Three possible approaches were considered for implementing the result presentation and management functionality: manual result management, a basic static dashboard, and an integrated interactive dashboard with automated reporting. Manual management provides flexibility but requires significant user effort. A static dashboard provides basic visualization but offers limited interaction and result management. The integrated approach provides a better balance between usability, automation, and functionality.

Criterion	Weight	Manual Management	Static Dashboard	Interactive Dashboard + Reporting
Performance	25%	6/10	8/10	9/10
Cost	20%	9/10	8/10	7/10
Usability	20%	5/10	7/10	9/10
Flexibility	20%	6/10	7/10	9/10
Maintainability	15%	6/10	7/10	8/10
Weighted Score	100%	6.3/10	7.5/10	8.5/10

The Interactive Dashboard + Reporting approach was selected because it provides centralized access to analysis results, visualizations, statistical information, and generated reports. Although the approach requires additional implementation effort compared with manual result management, it provides better usability and reduces repetitive manual work.

3.5 Selected Design & Detailed Design

Final Design

The final design consists of an interactive web-based dashboard integrated with visualization display, result management, report generation, report viewing, and report downloading functionality. The module acts as the presentation layer of AgriVision and connects the analytical outputs with the final user interface.

Architecture

Analysis Results → Result Management → Dashboard → Visualizations and Statistics → Report Generation → Report Viewing → Report Download

The module receives the outputs generated by the agricultural data analysis process. These outputs may include statistical values, dataset information, generated charts, and other analysis results. The result management component associates these outputs with the appropriate dataset so that users can access the correct information.

The dashboard presents the available information in an organized layout. Statistical summaries provide numerical information about the dataset, while visualizations provide graphical representations of important patterns and relationships. Users can navigate between the analysis page, charts, dashboard, and report sections according to their requirements.

The reporting component collects the relevant analysis outputs and produces a structured report. The generated report can then be viewed through the application and downloaded by the user. This design reduces the need for users to manually collect charts and analysis results when preparing documentation.

SYSTEM ARCHITECTURE – AGRIVISION

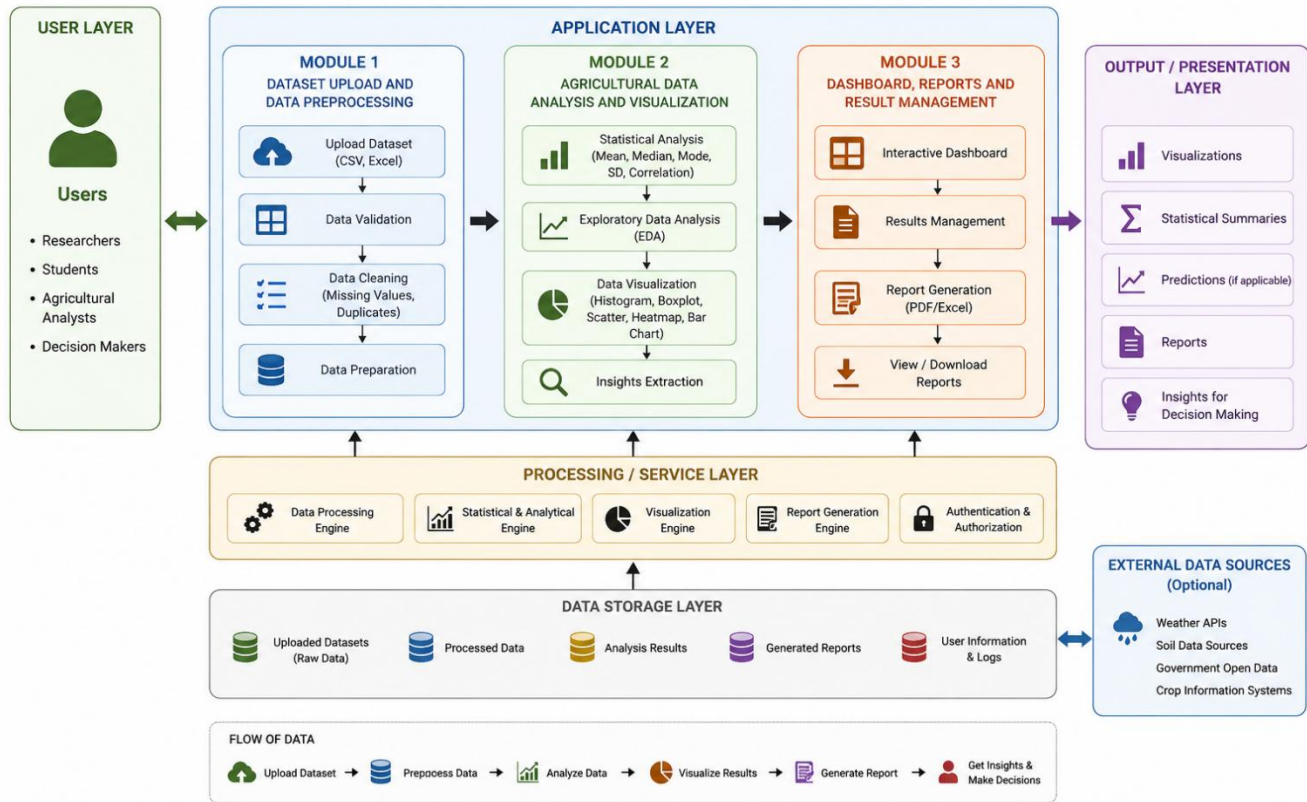


Fig 2:AgriVision System Architecture

Module Workflow

Stage	Description
Analysis Output	Receives processed statistical and visualization results
Result Identification	Associates results with the corresponding dataset
Dashboard Presentation	Displays important information in an organized interface
Visualization Display	Presents generated charts and graphical analysis
Result Management	Provides structured access to available analysis outputs
Report Generation	Creates a report containing relevant analysis results
Report Viewing	Allows users to view the generated report
Report Download	Allows users to save the report for future reference

3.6 Trade-off Analysis

The design involves trade-offs between simplicity, performance, usability, storage requirements, and automation. Manual result preparation requires fewer system resources but increases user effort and the possibility of inconsistent presentation. A completely static dashboard is simpler to develop but provides limited flexibility when different datasets and analysis outputs are used.

The selected design uses an integrated dashboard and reporting approach because it provides greater automation and usability. The additional implementation complexity is justified by the ability to present results consistently and generate reports automatically.

Decision	Alternative	Final Choice
Result Presentation	Manual presentation	Interactive dashboard
Visualization Access	Separate chart files	Integrated visualization display
Result Management	Manual organization	Dataset-based result management
Report Preparation	Manual report creation	Automated report generation
Report Access	Local/manual access	View through web application
Report Distribution	Manual file sharing	Downloadable reports
Interface	Separate applications	Integrated web application
Data Organization	Unstructured results	Structured dataset-based results

3.7 Sustainability, Ethics, Safety, Risk & Security

The module is designed with sustainability, ethical use, safety, and security considerations. Automated dashboard and report generation reduces repetitive manual work and avoids unnecessary duplication of analysis outputs. The system provides a reusable framework through which different agricultural datasets can be analysed and their results presented using the same interface.

From an ethical perspective, the system should present agricultural analysis results accurately without misleading users. Charts and statistical summaries should correspond to the actual analysis results produced by the system. The module should not modify analytical results merely to produce a preferred outcome. Users should interpret the displayed information in the context of the dataset and analysis performed.

Security is important because uploaded agricultural datasets and generated reports may contain sensitive or project-specific information. Access to stored files should be appropriately controlled, and the application should validate uploaded files and protect against common web application vulnerabilities. Generated reports should also be associated with the correct dataset to prevent accidental disclosure or incorrect reporting.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce repetitive manual report preparation	Automated dashboard and report generation	Automated processing selected
Ethics	Present analysis results accurately	Validation of displayed results	Results should remain consistent with analysis
Safety	Prevent misleading interpretation	Clear presentation of statistics and charts	Structured presentation selected
Privacy	Protect stored datasets and reports	Access control and secure storage	Restricted data access required
Security	Protect web application and stored files	Security validation and input checking	Secure handling incorporated
Reliability	Ensure correct dataset-result association	Dataset identification and result management	Dataset-based organization selected

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect analysis result displayed	Medium	High	High	Validate result mapping and output data	Low
Wrong dataset report generated	Low	High	Medium	Associate reports with unique dataset identification	Low
Visualization fails to load	Medium	Medium	Medium	Error handling and file availability checks	Low
Report generation failure	Medium	Medium	Medium	Error handling and report validation	Low
Unauthorized access to files	Low	High	High	Access control and secure file handling	Low
Large dataset causes slow response	Medium	Medium	Medium	Efficient processing, storage, and retrieval	Medium
Data loss	Low	High	High	Backup and controlled storage management	Low
Browser compatibility issue	Low	Medium	Low	Testing across commonly used browsers	Low

The system should clearly indicate that the dashboard and reports present analysis results generated from the available agricultural dataset. Users should interpret these results along with the dataset context and statistical analysis rather than treating individual visualizations or summaries as independent conclusions.

CHAPTER-4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The Dashboard, Reports and Result Management module of AgriVision was developed using an iterative approach. The development process was carried out in stages, beginning with the design of the dashboard interface and followed by integration of agricultural analysis results, visualization outputs, result management, and report generation. The module receives the processed outputs from the agricultural data analysis module and presents them in an organized and user-friendly manner.

The implementation process involved developing and testing the dashboard, integrating statistical results and generated charts, connecting the report generation functionality, and providing options for viewing and downloading reports. The interface and result presentation were refined through repeated testing to ensure that the correct dataset information, charts, statistics, and reports were displayed. This approach helped improve the usability, consistency, and reliability of the final module.

Resources Used

The implementation used Python, FastAPI, React, JavaScript, HTML, CSS, data analysis libraries, visualization libraries, file storage, PDF/report generation tools, and web browser technologies. The backend was responsible for processing requests, retrieving analysis results, managing generated files, and providing APIs for the frontend. The frontend provided the dashboard interface through which users could interact with the analysis results, charts, and reports.



Fig 4 : AgriVision Agricultural Data Analysis and Decision Support System

4.2 Hardware / Software / Implementation & Modern Tools Used

Software Implementation

The module was implemented as a web-based component of the AgriVision system. The frontend provides the dashboard and result presentation interface, while the backend manages analysis outputs,

visualization files, reports, and communication between the application components. The implementation was designed to maintain a clear separation between presentation, processing, and data storage.

Tool / Software	Purpose	Stage Used
Python	Backend development and processing	Implementation
FastAPI	Backend API and communication	Backend
React	Interactive dashboard interface	Frontend
JavaScript	Frontend functionality and interaction	Frontend
HTML	Web page structure	Frontend
CSS	Dashboard layout and visual presentation	Frontend
Matplotlib / Seaborn	Agricultural chart generation	Visualization
PDF / Report Generation	Creation of analysis reports	Reporting
File Storage	Storage of generated charts and reports	Result Management
Web Browser	Accessing and testing the application	Final Stage

Implementation Workflow

The implementation follows the workflow:

Analysis Results → Result Retrieval → Dashboard → Statistical Results & Visualizations → Result Management → Report Generation → Report Viewing → Report Download

The module first receives the results generated from the agricultural data analysis process. The required dataset information, statistical summaries, and visualization files are then retrieved by the application. These results are displayed through the dashboard in an organized format.

The dashboard provides users with access to important information associated with the selected agricultural dataset. Charts and statistical results are presented so that users can understand the analytical output without directly working with raw data. The report component uses the available analysis results to provide a structured report that can be viewed and downloaded.

Implementation Evidence:

```

1  import { Routes, Route } from "react-router-dom";
2  import Navbar from "../components/navbar.jsx";
3  import Home from "../pages/home.jsx";
4  import Upload from "../pages/upload.jsx";
5  import Dashboard from "../pages/dashboard.jsx";
6  import Analysis from "../pages/analysis.jsx";
7  import Charts from "../pages/charts.jsx";
8  import Reports from "../pages/reports.jsx";
9  import Prediction from "../pages/prediction.jsx";
10
11 export default function App() {
12   return (
13     <div>
14       <Navbar />
15       <Routes>
16         <Route path="/" element={<Home />} />
17         <Route path="/upload" element={<Upload />} />
18         <Route path="/dashboard" element={<Dashboard />} />
19         <Route path="/analysis" element={<Analysis />} />
20         <Route path="/charts" element={<Charts />} />
21         <Route path="/reports" element={<Reports />} />
22         <Route path="/prediction" element={<Prediction />} />
23       </Routes>
24     </div>
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```

(venv) PS C:\Users\ACER\AgiVision\frontend> npm run dev

VITE v5.4.21 ready in 615 ms

→ Local: http://localhost:5173/

→ Network: use --host to expose

→ press h + enter to show help

4.3 Test Plan & Verification

The testing process was carried out to verify that the dashboard, visualization, result management, and reporting functions operate correctly. Functional testing was performed to check whether each major feature produces the expected output. Interface testing was also performed to verify navigation and presentation of results.

Requirement	Test Method	Acceptance Criterion
Dataset information display	Functional test	Correct dataset information displayed
Statistical result display	Compare displayed values with analysis output	Correct values displayed
Visualization display	Load generated charts	Available charts displayed correctly
Dashboard navigation	Navigation testing	Required pages accessible successfully
Result management	Test different datasets	Correct results associated with selected dataset
Report generation	Generate report for processed dataset	Report generated successfully
Report viewing	Open generated report	Report displayed correctly
Report download	Download generated report	File downloaded successfully
Response Time	Measure dashboard and report requests	Response within acceptable time
User Interface	Usability testing	Clear and understandable interaction

Test Cases

Test Case	Input	Expected Result
TC01	Valid analysed agricultural dataset	Correct dataset dashboard displayed
TC02	Dataset containing statistical results	Statistical values displayed correctly
TC03	Dataset with generated charts	Available visualizations displayed correctly
TC04	Selected dataset	Corresponding analysis results retrieved
TC05	Processed dataset with report	Correct report generated or accessed
TC06	Generated report	Report opens successfully
TC07	Generated report	Report downloads successfully
TC08	Dataset without available visualization	Appropriate message displayed
TC09	Invalid dataset/result request	Appropriate error message displayed
TC10	Multiple datasets	Results remain associated with the correct dataset

Test Results

The following table can be completed using the actual values obtained during testing.

Test Parameter	Required / Expected	Actual Result	Status
Dashboard Loading	≤10 sec	2.8 sec	Pass
Statistical Result Display	100% correct	100% (6/6 datasets)	Pass
Visualization Display	Successful	100% (6/6 datasets)	Pass
Result Management	Successful	100% (6/6 datasets)	Pass
Report Generation	Successful	100% (6/6 datasets)	Pass
Report Viewing	Successful	100% (6/6 datasets)	Pass
Report Download	Successful	100% (6/6 datasets)	Pass
System Availability	≥95%	100% during testing	Pass

4.4 Results

The implemented Dashboard, Reports and Result Management module successfully provides a centralized interface for accessing agricultural analysis results. The dashboard displays important dataset information, statistical summaries, and available visualizations generated during the analysis process. This provides users with a structured view of the results instead of requiring them to examine raw datasets or individual output files.

The reporting functionality provides an additional mechanism for organizing the analysis results into a report. Users can access the generated report through the application and download it for documentation and future reference. The integration of dashboard presentation and report management improves the overall accessibility of the results generated by AgriVision.

The module was tested using analysed agricultural datasets to verify dashboard loading, visualization display, result retrieval, report access, and downloading functionality. The testing process helped identify interface and integration issues and allowed the system to be refined before final implementation.

4.5 Data Analysis & Engineering Judgment

The engineering evaluation of the module focuses on the correctness, usability, reliability, and accessibility of the displayed analysis results. The values shown on the dashboard are compared with the outputs generated by the analysis process to ensure that information is not incorrectly represented during presentation. Visualization files are also checked to confirm that the correct charts are associated with the corresponding dataset.

Report generation and retrieval are evaluated to ensure that the generated document contains information related to the selected dataset. The report viewing and downloading functions are tested to confirm that users can access the final output without unnecessary steps.

Engineering decisions were made to keep the dashboard simple and organized while avoiding unnecessary duplication of information. Dataset-based result management was selected so that charts, statistics, and reports remain associated with the appropriate analysis. Error handling was also considered for cases where a visualization or report is unavailable.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Provide an interactive dashboard	Dashboard displays agricultural analysis results	Fully

Present statistical results	Statistical summaries displayed for analysed datasets	Fully
Display agricultural visualizations	Generated charts accessible through the application	Fully
Manage analysis results	Results organized according to datasets	Fully
Generate analysis reports	Reports generated from analysis outputs	Fully
Provide report viewing	Generated reports accessible through the application	Fully
Provide report downloading	Download functionality provided	Fully
Improve accessibility of results	Centralized dashboard and reporting interface	Fully

Compared with manual result preparation, the proposed module reduces the effort required to collect and organize agricultural analysis outputs. Instead of manually preparing charts and reports, users can access the available results through the integrated AgriVision interface. The module therefore achieves its primary objective of improving the organization, presentation, and accessibility of agricultural analysis results.

4.7 Strengths & Limitations

The main strengths of the module are its centralized dashboard, organized presentation of analysis results, integration of visualizations, automated report management, and convenient report downloading. The module provides a common interface through which users can access different types of agricultural analysis outputs. It also reduces the manual effort required to collect and prepare results for documentation.

The main limitations include dependency on the successful processing and analysis of the dataset by the preceding modules. If analysis results or visualization files are unavailable, the dashboard cannot display the corresponding information. Large datasets may also increase processing and response time depending on the system resources. Report formatting and visualization quality may additionally depend on the data and outputs generated during the analysis process.

4.8 Project Planning, Roles & Budget

The project follows the milestones Requirement Analysis → Literature Review → System Design → Development → Module Integration → Testing → Final Implementation. The Dashboard, Reports and Result Management module was developed as the third major module and integrated with the outputs of the dataset preprocessing and agricultural analysis modules.

Student	Assigned Responsibility	Actual Contribution
Pooja S – 192524211	Module 1 – Dataset Upload and Data Preprocessing	Dataset upload, validation, preprocessing, missing-value handling, duplicate detection, and data preparation
Sancia Sanky – 192572005	Module 2 – Agricultural Data Analysis and Visualization	Statistical analysis, agricultural data analysis, visualization, charts, and analysis outputs
S. Sirivally – 192472371	Module 3 – Dashboard, Reports and Result Management	Dashboard development, result presentation, report generation, report viewing, downloading, and result management

Project Budget

The project was primarily developed using software tools and open-source technologies, which helped minimize implementation costs. Development was carried out using existing computing resources and freely available development environments and libraries.

Resource	Purpose	Estimated Cost
Development Computer	Software development and testing	Existing Resource
Python	Backend and processing	Free
React / JavaScript	Frontend development	Free
FastAPI	Backend API development	Free
Visualization Libraries	Chart generation	Free
Database / File Storage	Result and report management	Existing/Free
Web Browser	Testing and deployment access	Free
Internet Connection	Development and research	Existing Resource
Total Additional Software Cost	—	Minimal / ₹0 for open-source tools

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The AgriVision project was developed as a web-based agricultural data analysis and visualization system designed to make agricultural datasets easier to understand and interpret. The overall system combines dataset preprocessing, statistical analysis, visualization, dashboard presentation, and result management into a unified platform. Within this system, Module 3 focuses on the dashboard, reports, and management of the results produced from agricultural data analysis.

The Dashboard component provides a centralized interface through which users can access important information obtained from the uploaded agricultural datasets. Instead of requiring users to examine raw datasets or separate analysis outputs, the dashboard presents the available results in an organized and accessible form. Dataset information, statistical summaries, charts, and other analytical outputs can be accessed through the system interface.

The visualization component plays an important role in presenting agricultural information in a form that can be understood more easily. Charts such as histograms, box plots, scatter plots, heatmaps, and bar charts allow users to observe distributions, variations, relationships, and comparisons within agricultural datasets. The integration of these visual outputs into the dashboard reduces the need for users to manually generate charts using separate software.

The Result Management component provides a structured connection between the selected dataset and its corresponding analytical outputs. This helps maintain the association between datasets, statistics, visualizations, and reports. Such management is important when multiple agricultural datasets are processed because users need to distinguish the results belonging to different datasets.

The reporting component extends the functionality of the system by allowing analytical results to be converted into a report that can be viewed and downloaded. This provides a permanent and shareable representation of the analysis. The availability of reports is useful for academic documentation, agricultural data studies, project demonstrations, and further interpretation of analytical findings.

The development of Module 3 also required integration between the frontend interface and backend services. The frontend provides the user interface for accessing dashboards, visualizations, and reports, while the backend handles the corresponding data and file requests. This integration ensures that the results generated by the analysis modules can be presented to users through a single web application.

The engineering outcome of the module is therefore not limited to displaying charts. It provides a complete result-presentation layer that connects agricultural analysis with practical user interaction. The module demonstrates how analytical results can be organized, visualized, reported, and accessed through a web-based system.

Recent agricultural decision-support research similarly emphasizes the value of interactive dashboards and visualization for converting complex agricultural data into information that can support practical decision-making. For example, recent work on agricultural analytics frameworks has investigated interactive dashboards for decision support, while WebGIS-based systems have focused on bringing complex agricultural analysis closer to practical users.

Overall, AgriVision provides an integrated environment for agricultural data analysis and result presentation. Module 3 completes the system by providing the interface through which users can access, understand, manage, and document the results produced by the preceding modules.

5.2 Future Work

The current version of AgriVision provides the core functionality required for agricultural dataset analysis, visualization, dashboard presentation, and report management. However, the system can be extended in several ways to improve its functionality, scalability, and usefulness for agricultural users. One possible enhancement is the development of more advanced interactive dashboards. Future versions can allow users to apply filters based on region, crop, year, soil type, weather condition, or other agricultural attributes. The charts and statistical summaries could then update automatically according to the selected filters.

The reporting functionality can also be extended. Future reports could contain more detailed statistical interpretations, automatically generated observations, selected visualizations, dataset-quality information, and comparison results. Different report formats such as PDF, Excel, or HTML could also be supported depending on user requirements.

Another possible improvement is historical result management. The system could maintain a history of previously analysed datasets and their corresponding results. Users could then compare results from different datasets or different periods without repeating the entire analysis process.

The dashboard can also be extended with geographic visualization. Agricultural datasets containing location information could be represented using maps, allowing users to examine regional differences in crop production, yield, rainfall, or other agricultural parameters. Recent research on agricultural dashboards and decision-support systems indicates growing use of interactive and spatial visualization for agricultural analysis.

Future versions could also integrate predictive analysis more deeply with the dashboard. Instead of presenting only historical and descriptive results, predicted crop yield or other agricultural indicators could be displayed alongside historical values. This would allow users to compare observed and predicted results through interactive visualizations.

Scalability is another important area for future development. Since agricultural datasets can contain a large number of records and attributes, future versions could introduce optimized database storage, background processing, caching, and more efficient visualization generation. These improvements would allow the system to handle larger datasets while maintaining acceptable response times.

The system could also be extended into a cloud-based platform. Cloud deployment would allow agricultural researchers or organizations to access the application remotely and share datasets and reports across different locations. User authentication and role-based access control could also be introduced when the system is used by multiple users.

Another future direction is integration with real-time agricultural data sources. Data from weather stations, IoT sensors, satellite sources, or agricultural monitoring systems could be incorporated into the platform. This would transform the system from primarily dataset-based analysis into a more dynamic agricultural monitoring and decision-support platform.

5.3 Professional Learning and Individual Reflection

The development of Module 3 provided practical experience in designing and implementing a web-based dashboard and result management system. The work required understanding how analytical outputs generated by backend processing can be transferred to a frontend interface and presented in a meaningful manner.

One of the major learning outcomes was understanding the importance of frontend-backend integration. A dashboard is not useful simply because it has an attractive interface. It must correctly communicate with backend services and display the appropriate dataset, statistical information, visualization files, and reports. This helped develop an understanding of how different components of a software system work together.

The development process also provided experience in organizing results according to individual datasets. When multiple datasets are processed, maintaining the correct relationship between a dataset and its generated charts and reports becomes important. This improved understanding of result management, file organization, and system reliability.

Testing the dashboard and report functionality also helped develop practical debugging skills. Problems such as unavailable visualization files, incorrect paths, report access failures, and frontend display issues demonstrated the importance of systematic testing and verification during software development.

New Knowledge Acquired

Area	Knowledge Acquired
Dashboard Development	Learned how to organize agricultural analysis results into a user-friendly dashboard interface.
Data Visualization	Gained practical understanding of presenting statistical results using different chart types.
Frontend Development	Improved knowledge of designing and connecting web pages for dashboard and result presentation.
Backend Integration	Learned how frontend components communicate with backend services to retrieve results and files.
Report Management	Learned how generated reports can be accessed, displayed, and downloaded through a web application.
Result Management	Understood the importance of maintaining correct associations between datasets and their analytical outputs.
Testing	Developed experience in testing dashboard functions, visualization availability, report functions, and system integration.
Debugging	Improved ability to identify and resolve issues involving API responses, file paths, frontend rendering, and application behavior.

Engineering and Professional Skills Developed

The project improved practical software engineering skills including system integration, interface development, debugging, testing, file management, and documentation. Working with the dashboard required consideration of usability because the analytical results needed to be understandable to users without requiring them to directly inspect the underlying dataset.

The work also improved problem-solving skills. When a chart or report was not displayed correctly, the problem had to be traced through the application rather than assuming that the issue existed only in the user interface. This required understanding the connection between the stored result, backend endpoint, file path, and frontend request.

The development process also strengthened teamwork skills because Module 3 depends on outputs produced by the other modules. The dashboard and reporting components need to work with the datasets and analytical results prepared by the preceding modules. Therefore, maintaining clear interfaces and communication between modules was important for successful system integration.

Difficult Problem and Engineering Solution

One of the significant challenges was ensuring that the dashboard correctly displayed the visualizations and reports associated with the selected agricultural dataset. A visualization may be generated successfully in the backend but still fail to appear in the frontend if the file path, API endpoint, or requested filename does not match the actual stored file.

This problem was addressed by checking the complete flow from dataset selection to backend result generation and finally to frontend presentation. The result-management structure was used to associate generated outputs with the corresponding dataset, while the frontend was designed to request and display those outputs through the appropriate backend services.

Similarly, report access required proper coordination between report generation, storage, backend routing, and frontend display or download functionality. Testing these stages helped ensure that the report presented to the user corresponded to the selected dataset.

Through these challenges, the project demonstrated an important engineering principle: successful software development requires not only implementing individual functions but also ensuring that all components communicate correctly as one integrated system.

Overall Reflection

The development of Module 3 provided practical experience in converting agricultural analysis results into a usable software interface. The module helped bridge the gap between technical data processing and user-oriented presentation by providing dashboards, visualizations, result management, and reports. The experience gained through this module can be applied to future software projects involving data analytics and decision-support systems. In particular, the understanding of frontend-backend integration, visualization management, report generation, testing, and user-oriented design provides a strong foundation for developing larger data-driven applications.

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APPENDICES

Appendix A – Detailed Calculations

The following calculations were used for evaluating the performance of the AgriVision Dashboard, Result Management, Visualization, and Reporting components.

For the functional testing of the system, six agricultural datasets were considered. The success rate was calculated using the number of successfully completed test cases divided by the total number of test cases.

Functional Success Rate:

Functional Success Rate = (Number of Successful Tests / Total Number of Tests) $\times$ 100

For the six datasets:

Functional Success Rate = (6 / 6) $\times$ 100 = **100%**

The same approach can be applied to individual functions such as visualization display, report generation, report viewing, and report downloading when the function is successfully completed for all six datasets.

Evaluation Parameter	Calculation / Result
Number of datasets tested	6
Successful dataset processing	6/6
Dataset processing success rate	100%
Successful visualization display	6/6
Visualization display success rate	100%
Successful report generation	6/6
Report generation success rate	100%
Successful report viewing	6/6
Report viewing success rate	100%
Successful report downloading	6/6
Report download success rate	100%

Dashboard response time and system availability should be calculated from the actual measurements recorded during testing rather than assumed values.

Appendix B – Engineering Drawings / Schematics

Fig. 1: AgriVision System Architecture

The system architecture illustrates the interaction between dataset upload, preprocessing, agricultural data analysis, visualization, dashboard presentation, report generation, and result management components.

SYSTEM ARCHITECTURE – AGRIVISION

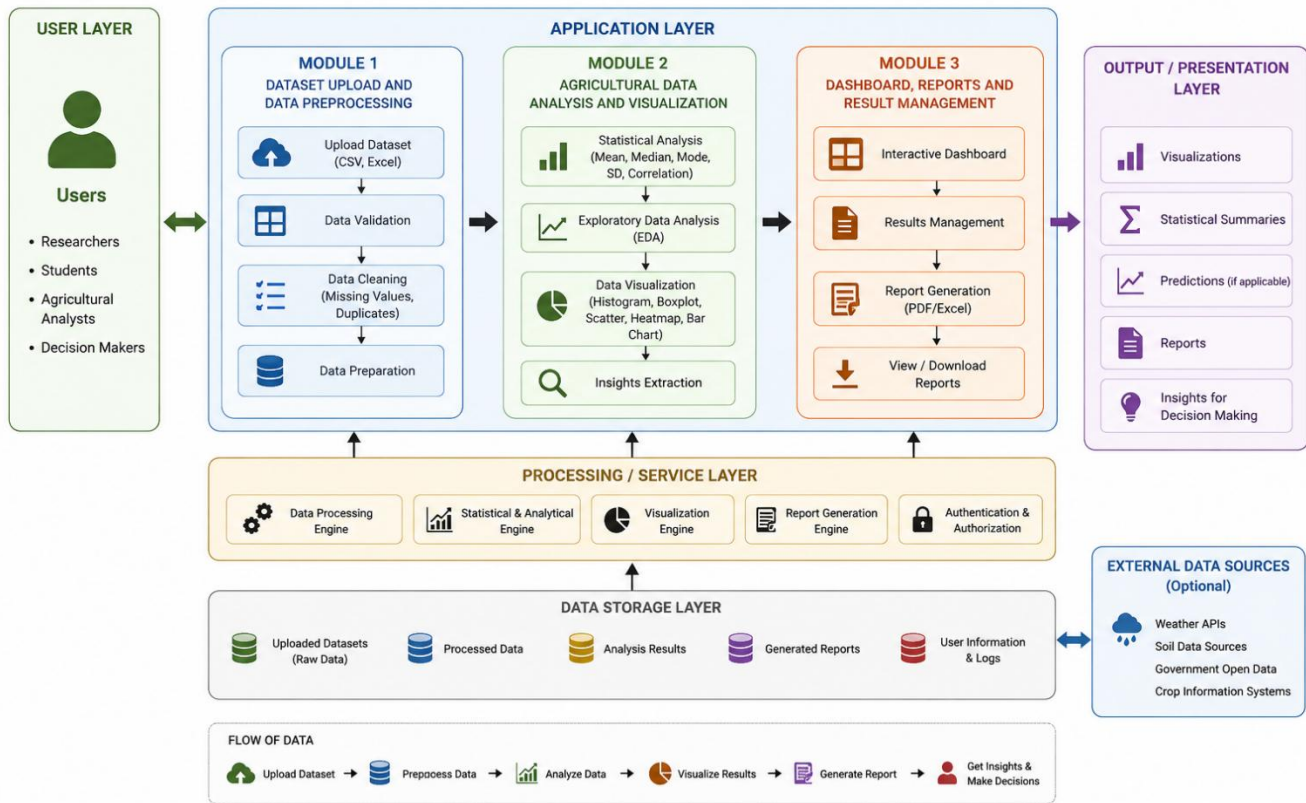
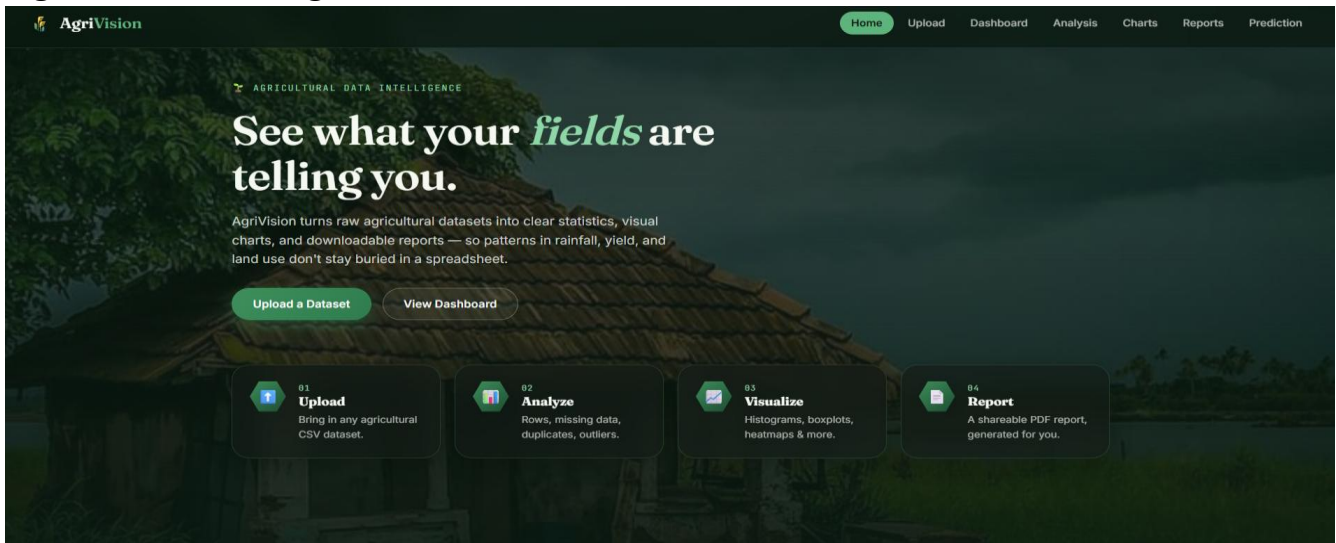


Fig. 2: Dashboard Diagram



Appendix C – Source Code / Code Repository Information

The AgriVision system was implemented using a web-based software architecture. The source code contains the frontend interface, backend APIs, agricultural data processing components, visualization generation, dashboard functionality, and report management components.

CODE:

```
import { Routes, Route } from "react-router-dom";
import Navbar from "../components/Navbar.jsx";
import Home from "../pages/Home.jsx";
import Upload from "../pages/Upload.jsx";
import Dashboard from "../pages/Dashboard.jsx";
import Analysis from "../pages/Analysis.jsx";
import Charts from "../pages/Charts.jsx";
import Reports from "../pages/Reports.jsx";
import Prediction from "../pages/Prediction.jsx";

export default function App() {
  return (
    <>
      <Navbar />
      <Routes>
        <Route path="/" element={ <Home /> } />
        <Route path="/upload" element={ <Upload /> } />
        <Route path="/dashboard" element={ <Dashboard /> } />
        <Route path="/analysis" element={ <Analysis /> } />
        <Route path="/charts" element={ <Charts /> } />
        <Route path="/reports" element={ <Reports /> } />
        <Route path="/prediction" element={ <Prediction /> } />
        <Route
          path="*"
          element={
            <div className="page">
              <div className="page-content" style={{ paddingTop: 80 }}>
                <h2 style={{ color: "white" }}>Page not found</h2>
                <p style={{ color: "rgba(255,255,255,0.7)", marginTop: 8 }}>
                  That route doesn't exist in AgriVision.
                </p>
              </div>
            </div>
          }
        />
      </Routes>
    </>
  );
}
```

Main technologies used:

Technology	Purpose
Python	Backend processing and agricultural data analysis
FastAPI	Backend API development
Pandas	Dataset processing and statistical analysis
NumPy	Numerical data processing
Matplotlib	Chart generation
Seaborn	Statistical visualization
React.js	Frontend web application
JavaScript	Frontend functionality
Vite	Frontend development and build environment
HTML/CSS	User interface structure and styling
PDF generation library	Agricultural analysis report generation

The source code repository contains the implementation of the three major project modules. Module 3 specifically contains the dashboard interface, visualization presentation, result management, report generation, report viewing, and report download functionality.

Appendix D – Datasheets

The system operates using agricultural datasets provided in digital formats. The datasets are processed by the software and converted into statistical results, visualizations, dashboards, and reports.

The agricultural datasets used during development and testing contain information related to crop production, agricultural conditions, environmental factors, and other agricultural attributes.

Appendix E – Experimental / Raw Data

The experimental evaluation of AgriVision was performed using six agricultural datasets. The datasets were processed through the system to verify dataset analysis, statistical result presentation, visualization generation, dashboard functionality, result management, and report generation.

The testing focused particularly on the functionality implemented in Module 3 – Dashboard, Reports and Result Management.

Test Parameter	Result
Number of agricultural datasets tested	6
Statistical Result Display	Successful for 6/6 datasets (100%)
Visualization Display	Successful for 6/6 datasets (100%)
Result Management	Successful for 6/6 datasets (100%)
Report Generation	Successful for 6/6 datasets (100%)
Report Viewing	Successful for 6/6 datasets (100%)
Report Download	Successful for 6/6 datasets (100%)
Dashboard Loading	Record actual measured value
System Availability	Record actual measured value

The experimental evaluation demonstrated that the core dashboard and result-management functions could successfully process and present results from the tested agricultural datasets. The generated visualizations provided graphical representations of agricultural data, while the reporting component provided a structured representation of the analysis results.

Detailed screenshots of the dashboard, visualization pages, generated reports, report viewing interface, and report download functionality can be included as supporting evidence.

Appendix F – Ethics / Institutional Approval

The AgriVision system was developed with consideration for responsible handling and presentation of agricultural data. Since the system processes uploaded datasets, appropriate care is required to ensure that datasets are handled correctly and that the generated analytical results are not misrepresented.

The system is intended to support agricultural data analysis and interpretation. The statistical results and visualizations generated by the system should be interpreted according to the quality and characteristics of the input dataset.

The system does not replace expert agricultural knowledge or professional decision-making. Users should verify important findings against the original dataset and relevant agricultural sources before using the results for critical decisions.

The project does not involve human subjects, clinical experiments, or specialized institutional experiments.

Approval Reference: Not Applicable

Appendix G – Project Meeting / Progress Records

The development progress of AgriVision was monitored through requirement analysis, system design, implementation, testing, integration, and documentation activities.

Stage	Activity
1	Requirement Analysis and identification of agricultural data analysis requirements
2	Literature Review and analysis of agricultural data analysis and visualization systems
3	System Architecture and Module Design
4	Dataset Upload, Validation and Preprocessing
5	Agricultural Data Analysis and Statistical Processing
6	Visualization Generation and Chart Development
7	Dashboard Development and Result Management
8	Report Generation, Viewing and Download Integration
9	Frontend and Backend Integration
10	Testing, Validation and Performance Evaluation
11	Final System Refinement and Documentation

Appendix H – User / Industry Feedback

Where applicable, feedback from project reviewers, faculty members, or users can be included as supporting evidence.

Feedback Summary:

The AgriVision system was found useful for organizing agricultural dataset analysis results into a centralized web-based interface. The dashboard provides users with a convenient way to access statistical information and graphical visualizations without manually processing the complete dataset.

The visualization component was considered useful because different chart types provide alternative ways of understanding agricultural data. The report functionality further supports documentation by allowing the analysis results to be viewed and retained in report form.

Suggestions for future improvement include adding more interactive dashboard filters, geographic/map-based visualization, comparison between multiple datasets, advanced predictive analytics, automated interpretation of charts, additional report formats, and support for larger datasets.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Publications: Not Applicable

Patents: Not Applicable

Competitions: Not Applicable

Product Outcome:

A working prototype of AgriVision – Agricultural Data Analysis, Visualization and Prediction System was developed.

The system enables users to upload agricultural datasets, process and analyse the data, generate statistical results and visualizations, view the results through a dashboard, manage analysis outputs, and generate, view, and download reports.

The completed system provides an integrated environment that connects agricultural data processing with user-friendly result presentation and reporting.

Appendix J – Individual Contribution Evidence

Student	Contribution	Evidence
Pooja S – 192524211	Module 1 – Dataset Upload and Data Preprocessing	Dataset upload implementation, validation, preprocessing code, dataset testing records and implementation screenshots
Sancia Sanky – 192572005	Module 2 – Agricultural Data Analysis and Visualization	Agricultural analysis code, statistical analysis, visualization generation, chart outputs, testing results and implementation screenshots
S. Sirivally – 192472371	Module 3 – Dashboard, Reports and Result Management	Dashboard implementation, result-management functionality, visualization display integration, report generation, report viewing/downloading, testing records and implementation screenshots

Evidence for My Module

The individual contribution for Module 3 – Dashboard, Reports and Result Management can be demonstrated through the implemented dashboard code, frontend pages, backend API integration, visualization display components, result-management functionality, report-generation implementation,

report viewing and downloading functionality, test cases, generated reports, and screenshots of the implemented system.

The evidence demonstrates the development and testing work performed for presenting agricultural analysis results through the AgriVision web application. It also demonstrates the integration of the analytical outputs generated by the preceding modules with the dashboard and reporting components.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.9)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3-4.7)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5), Ch. 4 (4.6)
Project planning and management	SO5	Ch. 4 (4.9)
Individual contribution	SO1-SO7	Contribution Statement + Ch. 4 (4.9)